

CS3 Rubric Case Study: Granger Causality & the U.S. Housing Market

Jonah Lee

DS 4002

Spring 2026

Instructor: Javier Rasero

TA: Ethan Meidinger

Submission Format

- Submit a link to your GitHub repository on Canvas
- Submit a PDF of your presentation slides on Canvas

Individual Assignment

Preparatory Assignments

Review the supplemental materials in the repository before beginning:

- Blog post: Introduction to Granger Causality (Aptech Systems)
- Technical article: Time Series Analysis Handbook, Chapter 4 - Granger Causality Test (PhD Students in Data Science, Asian Institute of Management)
- README.md in the project repository

This case study places you in the role of a data analyst at a housing policy research organization. You will apply Granger causality to real economic data to determine whether average hourly wages can improve forecasts of the U.S. National Home Price Index. Along the way you will practice the full data science workflow.

Course Learning Objectives:

- Use analysis and critical thinking skills to interpret time series results
- Prepare findings for presentation to your peers

What Am I Going To Do?

Read the Hook Document to understand the scenario. Then review the supplemental materials and README. Using the FRED datasets provided or linked in the repository, you will:

- Download and merge the average hourly wage and USNHPI time series
- Perform exploratory data analysis (EDA) to understand trends
- Test both series for stationarity using ADF and KPSS tests
- Fit a Vector Autoregression (VAR) model, selecting lag order using AIC/BIC/HQIC/FPE
- Run the Granger causality test and interpret the resulting p-values
- Communicate your findings and their policy implications in a presentation

Deliverables:

- GitHub Repository: all materials needed to reproduce your results
- Presentation: a slide deck guiding the audience from problem framing to conclusions

How Will I Know I Have Succeeded?

You will meet expectations on your submission when you follow all of the criteria in the rubric table below.

Rubric

Spec Category	Spec Details
Formatting	Repository • Submit a link to your GitHub repo on Canvas • Everything must be contained in or linked from the repo • Required folders: README.md, LICENSE.md, SCRIPTS/, DATA/, OUTPUT/ • Presentation PDF included in repo root Presentation • 10 to 12 slides in PDF format • Use PDF format for all documents; appropriate formats for code and data
Github Repository	Goal: A professional, reproducible record of your analysis. A reviewer should be able to clone your repo and fully recreate your results. README.md • Explain how to use the repo with section headers • Include software requirements, Python version, and library versions • Include step-by-step instructions for reproducing the analysis and a map of the repo structure LICENSE.md • Choose an appropriate license (MIT is acceptable) SCRIPTS/ folder • Name scripts in the order they should be run to reproduce results • Include header comments describing purpose, inputs, and outputs • Include clear inline comments explaining each major step DATA/ folder • Include raw and cleaned datasets (downloaded from FRED) • If data must be downloaded, include a clear instruction file with source URLs OUTPUT/ folder

	• Include all figures and tables used in the presentation Use informative, descriptive filenames
Presentation	Goal: Guide your audience. Your deck must include the following slides: Title and Outline (1 slide) • State your main takeaway in headline format; provide an overview of the presentation flow Motivation, Context, and Research Question (1-2 slides) • Explain why housing affordability matters and why wages are a natural predictor candidate ◦ State the research question clearly Data Explanation and Acquisition (1 slide) • Describe both FRED time series: date range, frequency, key variables • Note any data merging, cleaning, or lag column construction you performed Exploratory Data Analysis (1 slide) • Present at least two EDA visualizations • Highlight key economic events visible in the data (2008 crisis, COVID-19 era, etc.) Analysis Plan and Justification (1 slide) • Present your analysis pipeline (stationarity testing → VAR order selection → Granger test) • Explain why Granger causality is appropriate for this data. Stationarity Testing (1 slide) • Show results of ADF and KPSS tests for both series • Explain what stationarity means and why it matters for this analysis • Describe any transformations applied Tricky Analysis Decision (1 slide) • Describe one meaningful judgment call • Explain why it mattered and how it affected your results Results and Conclusions (1 slide) • Present the Granger causation matrix and p-values clearly • State your conclusion and answer the research question • Distinguish between statistical predictability and real-world causation Limitations and Next Steps (1 slide) • Discuss what the model cannot tell us • Suggest extensions References and Acknowledgements (1 slide) Cite all datasets and libraries used

Materials

Goal: Provide everything a student needs to complete the analysis from scratch.

- Blog post: "Introduction to Granger Causality" - Aptech Systems
- Technical reading: Time Series Analysis Handbook, Chapter 4
- Raw CSV files from FRED: Average Hourly Earnings and S&P/Case-Shiller U.S. National Home Price Index
- Cleaned and merged dataset ready for analysis
- Annotated Python scripts